EdgeTrust-VANET: Hierarchical Trust-Adaptive Secure Communication Framework

Final Year Project (Capstone) — Comprehensive Progress & Engineering Roadmap

Project Domain: Connected Autonomous Vehicles (V2X) & Edge Security	**Repository:** edgetrust-core & edgetrust-ml	**Date:** October 2026
Simulation Stack: SUMO 1.22.0 + OMNeT++ 6.4.0 + Veins 5.2	**Edge AI Engine:** Random Forest & Logistic Regression (C++ Native)	**Status:** Phases 1 & 2 Complete (Phase 3 Active)

Executive Summary: EdgeTrust-VANET delivers a multi-tiered, defense-in-depth cybersecurity and dynamic trust management framework for Connected Vehicles. The architecture seamlessly integrates On-Board Unit (OBU) heuristic pre-filtering, Roadside Unit (RSU) native Edge AI classification, dynamic Bayesian-EMA trust scoring, and Inter-RSU (I2I) cooperative threat quarantine over an edge backbone. This document details all completed engineering milestones (including extensive ML pipeline benchmarks and C++ native deployments) and establishes a rigorous roadmap for remaining Fog-level aggregation, SDN adaptive forwarding, and multi-density stress testing.

1. Completed Work & Current Technical Achievements

1.1 Integrated Simulation Stack & Network Architecture (SUMO + OMNeT++ + Veins)

- **Full Bi-Directional Co-Simulation:** Successfully coupled Eclipse SUMO 1.22.0 (microscopic multi-lane traffic) with OMNeT++ 6.4.0 (discrete event network simulation) through Veins 5.2 TraCIScenarioManager over a robust TCP socket interface (Port 9999).
- **Physical & MAC Layer Conformance:** Deployed IEEE 802.11p / DSRC Wave wireless communication stack (10 MHz channel at 5.9 GHz, transmission power 20 mW, bitrate 6 Mbps, obstacle shadow fading, and receiver sensitivity -98 dBm).
- **Two-Grid Road Network Topology:** Engineered an interconnected dual-intersection road network featuring Grid 1 (Junction J13, SUMO [-18.0, 8.0]) and Grid 2 (Junction J20, SUMO [102.0, 8.0]) bridged by a 120.31 m bidirectional arterial corridor (E10/-E10) with realistic cross-grid and turning flows.
- **Strict 85 m DSRC Radio Range Boundary & Sector Assignment:** Enforced realistic 85 m communication limits at physical decider and application levels. Implemented proximity-based serving assignment: nodes closer to RSU 0 are designated [PRIMARY SERVING] by RSU 0 (Grid 1 Sector), while RSU 1 tracks them as [HANDOVER OBSERVER] inside the overlapping transition corridor (93 m - 143 m).

1.2 OBU-Based Local Heuristic Filtering (Vehicle-Tier First Line of Defense)

A 4-tier distributed heuristic filtering engine was implemented directly on vehicle On-Board Units (OBUs) to evaluate outgoing and incoming Basic Safety Messages (BSMs) in real time before forwarding:

Heuristic Tier	Evaluation Mechanism	Thresholds & Physical Boundaries	Attack Mitigated
Tier 1: Kinematic Plausibility	Checks physical limits of urban vehicular motion.	Speed: 0 to 45 m/s (162 km/h) Accel: -8.5 to +6.0 m/s² Max implied velocity: 55 m/s	Constant/Random Position Offsets, Position Mirroring, Out-of-bounds speed.
Tier 2: CAN Bus vs. BSM Cross-Check	Compares internal vehicle odometry against reported broadcast data.	Discrepancy tolerance: |V_can - V_bsm| ≤ 2.5 m/s Static moving check: v_can > 1.5 m/s & v_bsm == 0	Zero Speed Report, Feigned Braking, Spoofed Sensor Telemetry.
Tier 3: Directional & Heading Consistency	Calculates spatial displacement vector and compares against reported compass heading.	Heading deviation: |Heading - MotionAngle| ≤ 110° Active only when vehicle speed > 1.5 m/s	Reversed Heading, Erratic FDI maneuvers, Ghost lane jumps.
Tier 4: Rate & Comm Anomaly Filter	Sliding 1.0s window monitoring transmission frequency and message delays.	Max burst rate: ≤ 12 BSM/sec Latency ceiling: ≤ 500 ms	DoS Broadcast Flooding (16 Hz), Stale Data Replay, Time Delay attacks.

Impact: Obvious rogue messages are dropped locally on the OBU (0% transmitted to RSU), reducing edge network overhead by up to 68%.

1.3 Machine Learning Pipeline & Native C++ Embedded Predictors (edgetrust-ml)

- **Dataset Unification (23,970 Records):** Merged 9 official VeReMi simulation archives (17,470 records, 72.88%), EdgeTrust-VANET live extractions (5,000 records, 20.86%), and V-RADD routing attacks (1,500 records, 6.26%) into a standardized dataset.
- **Zero-Leakage Architectural Audit:** Conducted Pearson correlation, Spearman rank, and Mutual Information audits proving zero target leakage. Enforced strict spatiotemporal group-splitting by vehicle ID (node_id) into 70% Train (16,885), 15% Val (3,546), and 15% Holdout Test (3,539).
- **Benchmark of 17 Classification Architectures:** Evaluated across ensemble trees, gradient boosters, neural nets, linear, and non-parametric models:

Model Architecture	Accuracy	F1-Weighted	Precision	Recall	ROC-AUC	Edge Latency	Deployment Status
Random Forest (Champion)	90.99%	90.97%	90.98%	90.99%	90.59%	23.5 ms	**C++ Native Deployed**
Logistic Regression	85.74%	85.62%	86.10%	85.74%	88.42%	0.12 ms	**C++ Native Deployed**
CatBoost Classifier	80.69%	79.79%	82.72%	80.69%	90.77%	1.45 ms	Benchmarked
LightGBM	72.42%	69.32%	77.52%	72.42%	89.77%	0.54 ms	C++ Predictor Built
XGBoost	71.56%	68.23%	76.62%	71.56%	89.32%	0.28 ms	Benchmarked
Extra Trees Classifier	90.82%	90.79%	90.84%	90.82%	90.31%	18.2 ms	Benchmarked
MLP Neural Network	86.12%	85.98%	86.45%	86.12%	88.90%	1.10 ms	Benchmarked
Stacking Ensemble	91.15%	91.12%	91.20%	91.15%	90.85%	31.4 ms	Benchmarked

- **Native C++ Embedded Inference:** Rather than executing Python bindings during OMNeT++ simulations, we compiled the trained models directly into C++ (RandomForestPredictor.cc/.h with 50 compiled decision trees, and LogisticRegressionPredictor.cc/.h with StandardScaler scaling vectors), enabling real-time zero-copy edge classification.
- **Full-Stack Real-Time Web Dashboard:** Implemented a complete Flask/Chart.js interactive dashboard (dashboard/app.py) featuring single-query evaluation, 17-model arena benchmarking, and live telemetry inspection.

1.4 Implementation of 15+ VeReMi NextGen Attack Scenarios

Engineered autonomous attack behaviors in EdgeTrustVehicleApp.cc spanning all major VeReMi NextGen and VANET attack categories:

Scenario ID	Attack Name	Injected Anomaly Pattern	Primary Detection Mechanism
VeReMi 1	Constant Pos Offset	Position coordinates shifted by fixed +28.0 m offset.	CAN vs BSM & Kinematic jump bound
VeReMi 2	Random Pos Offset	Coordinates randomized between ±20 m and ±70 m.	Implied velocity violation (> 55 m/s)
VeReMi 3	Position Mirroring	Reflects vehicle position across road centerline into opposite lane.	Spatial lane boundary & heading check
VeReMi 4-5	Constant/Random Speed Offset	Claims speed offset by +12 m/s or ±6 m/s continuously.	Spatial displacement derivative vs reported v
VeReMi 6	Zero Speed Report	Claims 0.0 m/s while vehicle is traveling at arterial speed.	CAN bus speed cross-validation
VeReMi 7-8	Sudden Stop / Sudden Const Speed	Freezes reported position/speed while vehicle maneuvers.	Temporal delta-v & kinematic trajectory
VeReMi 9	Reversed Heading	Heading angle flipped 180° opposite to motion direction.	Motion trajectory vs compass angle
VeReMi 10-11	Feigned Braking / Accel Mult	False deceleration (-7 m/s²) or 3.5x acceleration multiplier.	Temporal consistency & calcAccel discrepancy
VeReMi 12	DoS Broadcast Flooding	Floods channel with 16 Hz high-frequency packet bursts.	Rate anomaly filter (sliding window > 12 Hz)
VeReMi 13	Traffic Congestion Sybil	Injects phantom ghost vehicle IDs to simulate fake gridlock.	Sybil identifier check & RSSI consistency
VeReMi 14-15	Data Replay & Time Delay	Replays stale beacons or delays timestamp by +3.0 seconds.	Timestamp freshness & latency penalty (>500ms)
VANET 16	Selective Blackhole Drop	Drops 70% of transit packets while maintaining BSM broadcasts.	Packet drop ratio (PDR) & comm score

1.5 14-Feature Extraction & Dynamic Multi-Tier Trust Engine

- **14 Observable Feature Vector:** Ingests [pos_x, pos_y, speed, heading, accel, pkt_sent, pkt_rcvd, pdr, latency, retx, rssi, trust_score, neighbor_trust_avg, hist_trust]. Streamed simultaneously to results/live_extracted_features.csv and synced with edgetrust-ml/data/.
- **Mathematical Trust Model:** Composite evidence combines 4 factors: $Evidence = 0.30 \cdot Consistency + 0.30 \cdot CommScore + 0.20 \cdot NeighborValidation + 0.20 \cdot Plausibility$. Updated via Exponential Moving Average (EMA): $Trust_t = \alpha \cdot Trust_{t-1} + (1-\alpha) \cdot Evidence$ with α **= 0.70**.
- **Hybrid Decision Matrix:** Fuses ML model classification and trust score into operational decisions: **ACCEPT** (Trust ≥ 0.70 & ML Normal), **WARN** (Trust < 0.70 or ML Malicious), **BLOCK** (Trust < 0.40 & ML Malicious).

1.6 Simulation Checkpointing & Inter-RSU (I2I) Cooperative Backbone

- **State Checkpointing & Warm-Start:** Automatically serializes full multi-tier trust states, kinematic records, attack indicators, and ML predictions into JSON (checkpoint_state_rsu0.json, checkpoint_state_rsu1.json). Simulations can be resumed instantaneously with pre-warmed trust states using the LiveDemo_FromCheckpoint configuration.
- **Periodic I2I Trust Synchronization:** RSUs exchange cooperative trust digests every 2.5s over an edge backbone link, aligning global reputation across Grid 1 and Grid 2, and pre-warming approaching vehicles before handover.
- **Urgent Threat Alerts & Preemptive Quarantine:** When an RSU detects a malicious node (BLOCK verdict), it immediately dispatches an urgent I2I threat alert over the edge backbone. The peer RSU pre-blacklists the rogue node before it physically crosses into the adjacent grid sector.
- **Canvas Visual Rendering:** Draws dynamic color-coded transmission arrows between vehicles and RSUs, along with a dashed I2I backbone line between RSU 0 and RSU 1 (cyan for periodic sync, magenta/red for urgent threat alerts).

2. Remaining Work & Technical Roadmap

While the foundational multi-RSU simulation, OBU heuristic pre-filtering, and embedded Edge AI detection are fully functional, the following modules represent the remaining research and development tasks required to complete the project:

2.1 Complete Inter-RSU Trust Synchronization Protocol

Formalize and harden the I2I consensus synchronization protocol. Implement bounded convergence proofs ensuring that trust scores converge within ≤ 3.0 seconds during vehicle handover between adjacent RSU cells. Add asynchronous heartbeats, ACK-based reliable backbone transport, and loss recovery over simulated fiber/Ethernet edge links.

2.2 Duplicate, Stale, and Conflicting Trust Update Resolution

In real-world decentralized VANETs, malicious nodes or asymmetric channel conditions produce conflicting trust reports. We must implement: (a) Lamport logical timestamps / sequence vectoring to discard stale or out-of-order trust packets, (b) Weighted Bayesian / Byzantine Fault Tolerant (BFT) consensus to reconcile conflicting neighbor validation scores, and (c) Sybil-resistant voting weights proportional to historical direct trust scores.

2.3 Fog-Based Regional Trust Aggregation Layer

Extend the architecture into a true 3-tier hierarchical framework: Vehicle (OBU) → Edge (RSU) → Regional Fog Server. The Fog layer will aggregate trust digests across clusters of 4-10 RSUs, maintain a persistent cross-district reputation ledger, detect distributed low-and-slow multi-intersection attacks, and coordinate city-wide Certificate Revocation List (CRL) distributions.

2.4 SDN-Based Adaptive Forwarding & Dynamic QoS Rerouting

Integrate Software-Defined VANET (SD-VANET) controller architecture (e.g., OpenFlow-compatible routing in OMNeT++). The controller will dynamically compute forwarding topologies based on live link trust scores, rerouting safety-critical BSM packets away from compromised or congested nodes, while dynamically provisioning bandwidth priority (QoS) for emergency advisories over non-safety traffic.

2.5 Multi-Density Traffic & Coordinated Multi-Vector Stress Testing

Perform comprehensive simulations across three standardized traffic density regimes: Low (15 veh/km), Medium (45 veh/km), and High / Congested (90-120 veh/km). Evaluate resilience against sophisticated simultaneous multi-vector attacks (e.g., combined DoS burst flooding + Sybil phantom nodes + gradual False Data Injection drift).

2.6 Comprehensive Empirical Evaluation & Benchmarking Suite

Conduct exhaustive comparative benchmarks across the entire metric spectrum: (a) Classification metrics: Accuracy, Precision, Recall, Macro/Weighted F1-score, False Alarm Rate (FAR), Missed Detection Rate (MDR), ROC-AUC; (b) Network performance metrics: End-to-End Latency, Packet Delivery Ratio (PDR), Jitter, Control Overhead, Handover Latency; (c) Computational efficiency: CPU load, memory footprint, single-packet inference latency under load.

2.7 Scalability Benchmarking on Large-Scale Topologies

Scale the simulation environment beyond the current two-RSU setup to multi-RSU arterial corridors (4 to 8 RSUs) and large-scale Manhattan urban grid topologies with 100+ concurrent vehicle trajectories to evaluate communication overhead and trust convergence scalability.

3. Project Status Matrix & Implementation Roadmap

Project Component	Key Technical Deliverables	Status	Target Phase
Simulation Stack Integration	SUMO 1.22 + OMNeT++ 6.4 + Veins 5.2 TraCI co-simulation, IEEE 802.11p DSRC PHY/MAC.	**COMPLETED**	Phase 1
OBU Heuristic Pre-Filtering	4-tier local heuristic engine (Kinematics, CAN cross-check, Heading, Burst rate).	**COMPLETED**	Phase 1
Machine Learning Pipeline	23,970 unified dataset, zero-leakage audit, 17 trained models, Web dashboard.	**COMPLETED**	Phase 1-2
Embedded C++ Predictors	Native C++ Random Forest & Logistic Regression compiled directly into RSU application.	**COMPLETED**	Phase 2
15+ VeReMi Attack Scenarios	Full VeReMi NextGen attack catalog + selective blackhole attack implemented in vehicles.	**COMPLETED**	Phase 2
14-Feature Live Extraction	Continuous feature streaming to CSV with automatic edgetrust-ml synchronization.	**COMPLETED**	Phase 2
Dynamic Trust Engine	Multi-tier Bayesian-EMA formulation (alpha=0.70), neighbor consensus, historical reputation.	**COMPLETED**	Phase 2
Simulation State Checkpointing	JSON state serialization, multi-RSU state persistence, warm-start simulation resumption.	**COMPLETED**	Phase 2
Two-Grid & Dual-RSU Scenario	120m corridor, 85m physical DSRC limits, primary serving vs handover observer assignment.	**COMPLETED**	Phase 2
Initial I2I Backbone Link	Periodic cooperative trust digest sync (2.5s) & urgent threat alert preemptive quarantine.	**COMPLETED**	Phase 2
Trust Sync Protocol Hardening	Formal convergence bounds, multi-hop propagation, ACK-based reliable edge transport.	**IN PROGRESS**	Phase 3 (Active)
Conflict & Byzantine Resolution	Lamport timestamps, out-of-order packet mitigation, Byzantine Fault Tolerant consensus.	**IN PROGRESS**	Phase 3
Fog-Based Regional Aggregation	Hierarchical OBU-RSU-Fog tier, cross-district global trust ledger, regional CRL dispatch.	**PLANNED**	Phase 4
SDN Adaptive Forwarding	OpenFlow SD-VANET controller, dynamic trust-based routing, QoS priority queuing.	**PLANNED**	Phase 4
Multi-Density Stress Testing	Low, Medium, and High traffic density regimes + simultaneous coordinated multi-vector attacks.	**PLANNED**	Phase 4
Full Benchmark Evaluation	Accuracy, F1, FAR, MDR, PDR, End-to-end latency, Handover delay, Hardware profiling.	**PLANNED**	Phase 5
Large-Scale Scalability Tests	4 to 8 RSU arterial corridor, Manhattan grid scaling, 100+ concurrent vehicles.	**PLANNED**	Phase 5

Summary & Next Actions: The EdgeTrust-VANET framework currently possesses an operational simulation and detection pipeline with native edge intelligence and distributed vehicle pre-filtering. The immediate focus is concluding Phase 3 (protocol hardening and Byzantine conflict resolution), followed by Phase 4 (Fog layer and SDN routing) and Phase 5 (comprehensive multi-density empirical benchmarking for final thesis publication).